SVG/SVG+

INPHASE

Rugged Reliability in a Compact Form

Next-Generation Harmonics Mitigation System

- System Serviceability: 10-year parts availability
- Amperage Range: 50 to 200 Amps modules
- Architectural Design: Based on proven "ASTRA"
- System Efficiency: Greater than 98% power conversion
- Capacitor Technology: Utilizes advanced film capacitor technology

BUILT TOUGH, DESIGNED COMPACT

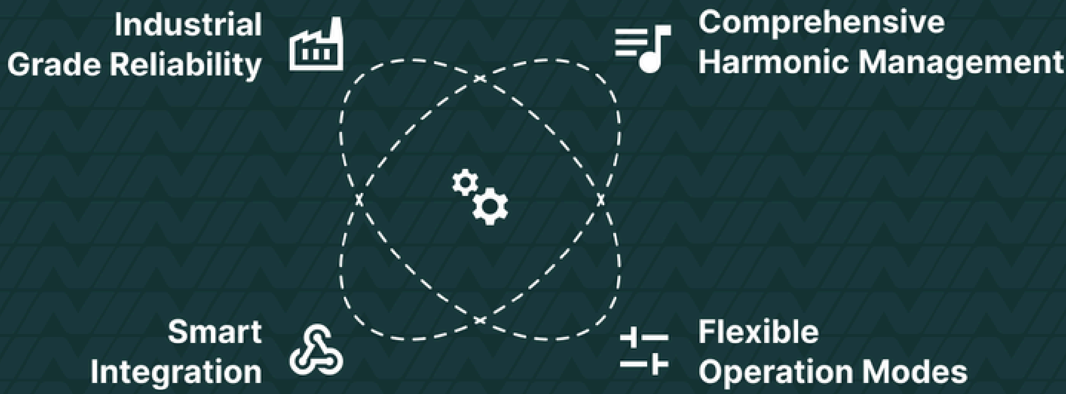
Registered Office Address: InPhase Power Technologies Private Limited, No. 59, Temple Road, Near Embassy Public School, Chikka Gollarahatti, Magadi Main Road, Bangalore - 560091

info@inphase.in **sales@inphase.in** **+91 96068 85500**

ENGINEERED FOR EXCELLENCE

Built on the proven ASTRA architecture, MicroBheem delivers industrial-grade reliability in a compact, modular design. Our Active Harmonic Filters represent the pinnacle of power quality management, offering comprehensive harmonic mitigation and reactive power compensation.

Advanced System Features Overview



KEY FEATURES

Comprehensive Harmonic Management

- Advanced harmonic compensation up to 50th order (AHF+)
- Up to 10 selectable frequencies
- Real-time adaptive filtering technology
- Superior harmonic mitigation performance

Flexible Operation Modes

- Pure Harmonic Mode: 100% capacity for harmonics
- Pure PF Mode: 100% capacity for reactive power
- Hybrid Mode: Optimized power quality management
- Load Balancing: 30% of rated capacity

Smart Integration

- Modular design for easy installation
- User-friendly interface
- Connect up to 32 units in parallel
- Compact footprint

Industrial-Grade Reliability

- Robust construction for harsh environments
- Advanced thermal management
- Comprehensive protection features
- Extended operational life

SPECIFICATIONS

DESCRIPTION	µBheem - AHF	µBheem - AHF+
Electrical Specification		
Ratings	25kVAR, 50kVAR, 70kVAR, 100kVAR, 150kVAR	25kVAR, 50kVAR, 70kVAR, 100kVAR, 150kVAR
Rated Voltage	415V, +10% / -15%	415V, +10% / -15%
Frequency	50Hz / 60Hz, +/-5%	50Hz / 60Hz, +/-5%
Connection Type	4Wire	4Wire
Topology	3-Level IGBT	3-Level IGBT
Efficiency	98%	98%
Control Basis	Closed loop operation	Closed loop operation
Harmonic range	up to 25th order harmonics	up to 50th order harmonics
Selectable Harmonics	10 selectable harmonics	20 selectable harmonics
Harmonic Attenuation	99% harmonics elimination	99% harmonics elimination
Reactive Time	0.1 ms	0.1 ms
Response Time	1 power cycle	1 power cycle
Operating Modes	Harmonics compensation only (H), PF compensation only (Q),Harmonics + PF compensation (H+Q), Fundamental current balancing	Harmonics compensation only (H)
Harmonics mode (H)	100% of rated capacity for harmonics	100% of rated capacity for harmonics
PF mode (Q)	100% of rated capacity for reactive power	-
Harmonics + PF mode (H+Q)	upto 50% of rated capacity for harmonics upto 50% of rated capacity for reactive power	-
Fundamental current balancing	Yes	Yes
Neutral Current Compensation	100% of rated capacity for <=ISOA AHF	100% of rated capacity for <=ISOA AHF
Reactive Power range	Both inductive and capacitive reactive power	-
CT Requirement	3Ph CTs,SA input,Class-1 or higher accuracy	3Ph CTs,SA input,Class-1 or higher accuracy
CT position	Grid / Source side	Grid / Source side
Parallel Expandability	up to 32 units in parallel. Any size unit combination possible.	up to 32 units in parallel. Any size unit combination possible.
Hybrid Operation Mode (Capacitor Bank Switching)	Yes	Yes
Installation Characteristics		
Installation Location	Indoor	Indoor
Construction Type	Modular	Modular
Panel Mounting configuration	Individual units have rack module design. Cabinet will be floor standing.	Individual units have rack module design. Cabinet will be floor standing.
Enclosure degree of protection	IP-20 (for module)	IP-20 (for module)
Cable Entry	Top/Bottom	Top/Bottom
Operating Altitude	<1000 m altitude	<1000 m altitude
Relative Humidity	0 - 95% non-condensing	0 - 95% non-condensing
Noise Level	<= 65db(A) @lm	<= 65db(A) @lm
Operating Temperature Range	Designed for 50° Operation	Designed for 50° Operation
Cooling configuration	Forced air cooling	Forced air cooling
Colour	RAL 7035	RAL 7035
Response Time	1 power cycle	1 power cycle
Module Dimensions (W x D x H)	590 x 685 x 335 in mm	590 x 685 x 335 in mm
Module Weight in kg	87kG / 95kG	87kG / 95kG
Communication		
Communications protocol	Modbus RTU	ModBus RTU
HMI	7-Inch Colour Touch Screen - HMI (Provided separately to be mounted in cabinet) Multiple modules can be connected to single HMI	Modular
SCADA	RS-485 port available for SCADA connectivity	Individual units have rack module design. Cabinet will be floor standing.
Display Parameters	All Power Parameters,Panel Parameters & Fault Log	IP-20 (for module)
Standards & Certification		
Design reference	EN/IEC 62477-1,IEC 61439-1	EN/IEC 62477-1,IEC 61439-1
EMC compliance	EN/IEC 61000-6-2,EN/IEC 61000-4.	EN/IEC 61000-6-2,EN/IEC 61000-4.
Product Certification	CE	CE

APPLICATIONS

MANUFACTURING

- Automotive plants
- Steel processing
- Textile industries
- Food processing units

INFRASTRUCTURE

- Data centers
- Hospitals
- Commercial buildings
- Metro stations

PROCESS INDUSTRIES

- Chemical plants
- Pharmaceutical facilities
- Paper mills
- Water treatment plants

Why Choose MicroBheem?

Proven Technology
Ensures tested and reliable power quality solutions across industries.

Local Expertise
Offers customized support and precision through local manufacturing and R&D.

Long-term Reliability
Guarantees durability and comprehensive after-sales support.

Future-ReadyDesign
- Scalable architecture
- Expandable capacity
- Industry 4.0 ready

Superior Performance
- Fast response time < 50μs
- High accuracy harmonic compensation
- Advanced digital control system
- Exceptional stability under varying loads

Cost-EffectiveOperation
- Reduced energy losses
- Lower maintenance requirements
- Extended equipment life
- Improved system efficiency

Explore Our Other PQ Products: AHF, AHF+

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